Building a Predictive Model for Diabetes Readmission

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1. Objective

Diabetes is a health condition that affects people across wide demographic groups, that can lead to severe health complications if not properly treated. Therefore, being able to diagnose and treat diabetes appropriately is crucial.

To that end, the objective for this assignment is to develop three (3) predictive models that can accurately identify diabetic encounters which will require readmission to inpatient care.

2. Data Preparation

For this assignment, a dataset was sourced from Kaggle, which captures more than 10,000 recorded cases and more than 50 features over 10 years (from 1998 to 2008) in the United States. These cases met the following criteria:

1. The case is an inpatient encounter.
2. The encounter involved any form of diabetes.
3. The term of stay lasted 1–14 days.
4. A lab test was performed as part of the encounter.
5. Medication was administered.

Dataset Source: kaggle.com/datasets/jimschacko/10-years-diabetes-dataset

After data exploration, some preprocessing was required to manage unknown values. Further, some feature engineering was done to define our target variable (will require readmission), amongst others. Lastly, some features were modified or removed due to limited usefulness in the final model as provided.

The dataset is also highly imbalanced: 88.84% of the patients were not readmitted to hospital. In consequence, we resampled the data to create a balanced training set.

3. Model Design

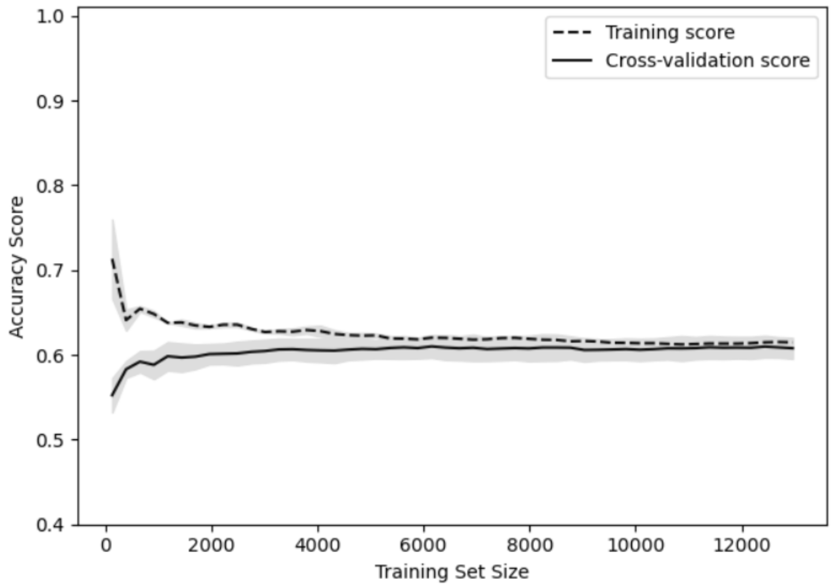
The three models used for this assignment are Logistic Regression (LR), Random Forest (RF), and Gradient Boosting (GBM). These three models were chosen due to the accuracy achieved by these models in comparison to others. This was determined by comparing the ROC curves and confusion matrices of various models after hyperparameter tuning. After tuning, each model displayed similar levels of accuracy and F1 score.

	LR	RF	GBM
Test Accuracy	58.5%	58.89%	58.36%
F1 Score	0.569	0.625	0.614

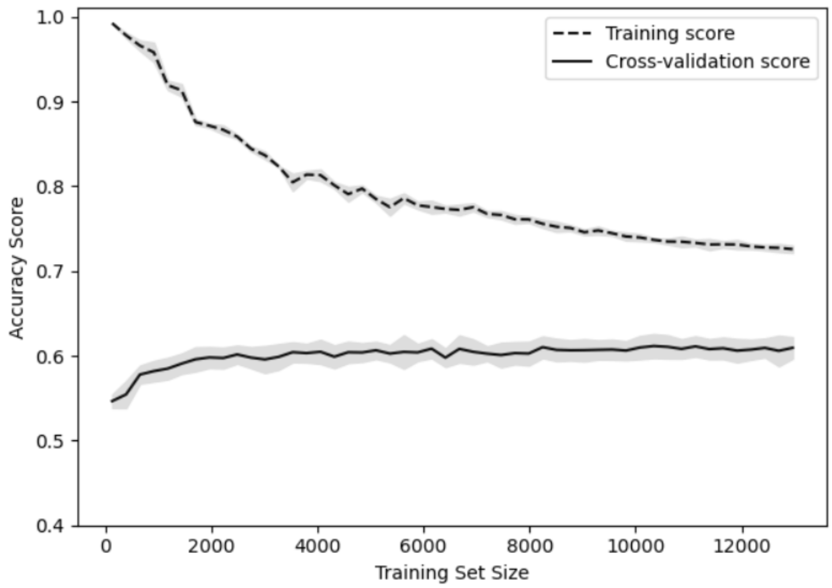
4. Model Evaluation

These three models were then compared using their respective learning curves (training score vs. cross-validation score across training set size) to identify which model had better performance.

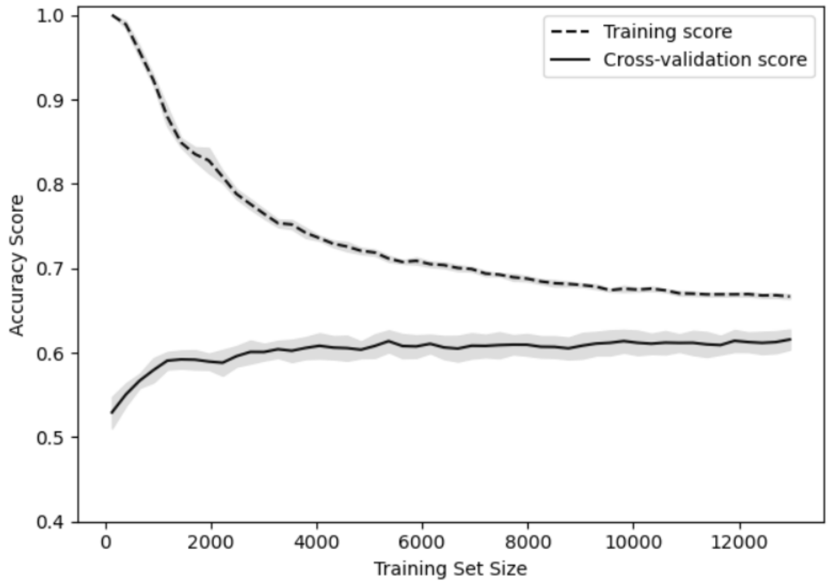
Logistic Regression (LR)



Random Forest (RF)



Gradient Boosting (GBM)



5. Conclusion

After evaluating the three models, we identified Gradient Boosting as the best performing model for our needs, identifying 58.36% of the readmission.

We also determined techniques that could have been done to improve the model, including sourcing data for a more balanced dataset or combining features during our feature engineering exercise, as opposed to dropping features due to low correlation with the target variable.